

Problem Set 3

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Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in R using the `incumbents_subset.csv` dataset. Include all of your code.

Loading in the dataset:

```
1 inc.sub <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2025/main/datasets/incumbents_subset.csv")
```

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

Checking density plots, we do not see a significant skew in spread that may benefit from a transformation.

```

1 plot(density(inc.sub$voteshare))
2 plot(density(inc.sub$difflog))
3 plot(density(inc.sub$presvote))

```

The first model regresses voteshare on difflog (the difference in campaign spending between the incumbent and the challenger).

```

1 reg1 <- lm(data= inc.sub, voteshare ~ difflog) #bivariate regression
2 summary(reg1)
3 stargazer(reg1,
4           type = "latex",
5           title = "Incumbent/Challenger Difference in Campaign Spending
6                   on Incumbent's Voteshare",
7           covariate.labels = "Difference Log Spending", #label
8           explanatory.vars
9           dep.var.labels = "Voteshare") #label predicted var

```

Table 1: Incumbent/Challenger Difference in Campaign Spending on Incumbent's Voteshare

	<i>Dependent variable:</i>
	Voteshare
Difference Log Spending	0.042*** (0.001)
Constant	0.579*** (0.002)
Observations	3,193
R ²	0.367
Adjusted R ²	0.367
Residual Std. Error	0.079 (df = 3191)
F Statistic	1,852.791*** (df = 1; 3191)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

```

1 pdf("PS03_reg1plot.pdf")
2 ggplot(inc.sub, aes(x = difflog, y = voteshare)) + #scatter plot using
3   geom_point(size = 0.7) + #point size smaller to avoid overlapping/
4   geom_smooth(method=lm) + #smooth method adds linear regression line
5   labs(x = "Difference Log Spending", y = "Voteshare") + #label axis

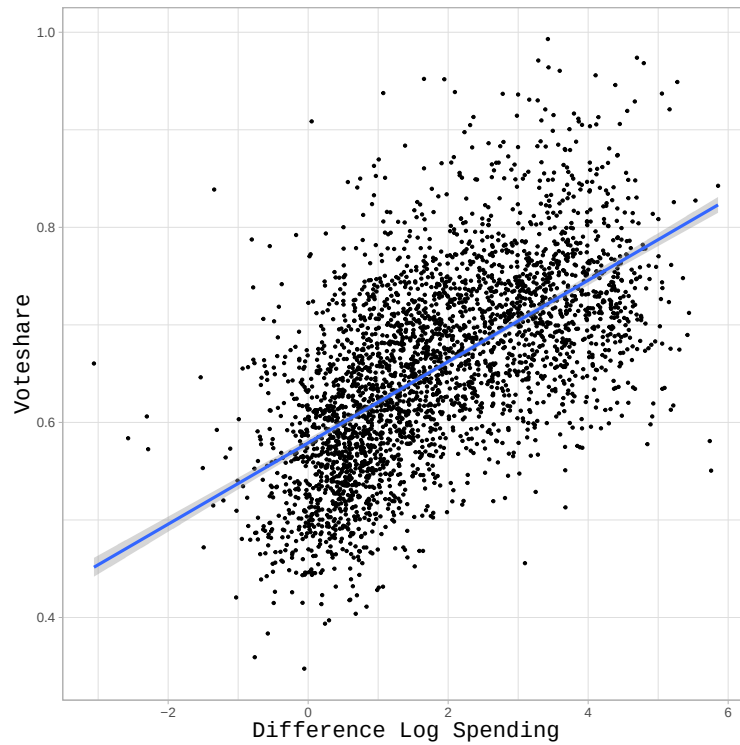
```

```

6 theme_light() + #change theme
7 theme(axis.title = element_text(size = 15, family = "mono")) #change
  font axis
8 dev.off()

```

Figure 1: Voteshare vs Difference Log



3. Save the residuals of the model in a separate object.

```

1 reg1_residuals <- reg1$residuals # save as new object res from reg1

```

4. Write the prediction equation.

$$\widehat{\text{voteshare}} = 0.579 + 0.042 \times \text{difflog} \quad (1)$$

On average, a one unit increase in difflog is associated with a 0.042 increase in voteshare.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
1 reg2 <- lm(data= inc.sub, presvote~difflog)
2 summary(reg2)
3 stargazer(reg2, #stargazer used for presenting model in table
4             type = "latex", #output code for use in latex
5             title = "Incumbent/Challenger Difference in Campaign Spending
6                   on the Presidential Candidate Voteshare of the Incumbent
7                   Party ",
7             covariate.labels = "Difference Log",
8             dep.var.labels = "Presidential Candidate Voteshare")
```

Table 2: Incumbent/Challenger Difference in Campaign Spending on the Presidential Candidate Voteshare of the Incumbent Party

	<i>Dependent variable:</i>
	Presidential Candidate Voteshare
Difference Log	0.024*** (0.001)
Constant	0.508*** (0.003)
Observations	3,193
R ²	0.088
Adjusted R ²	0.088
Residual Std. Error	0.110 (df = 3191)
F Statistic	307.715*** (df = 1; 3191)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

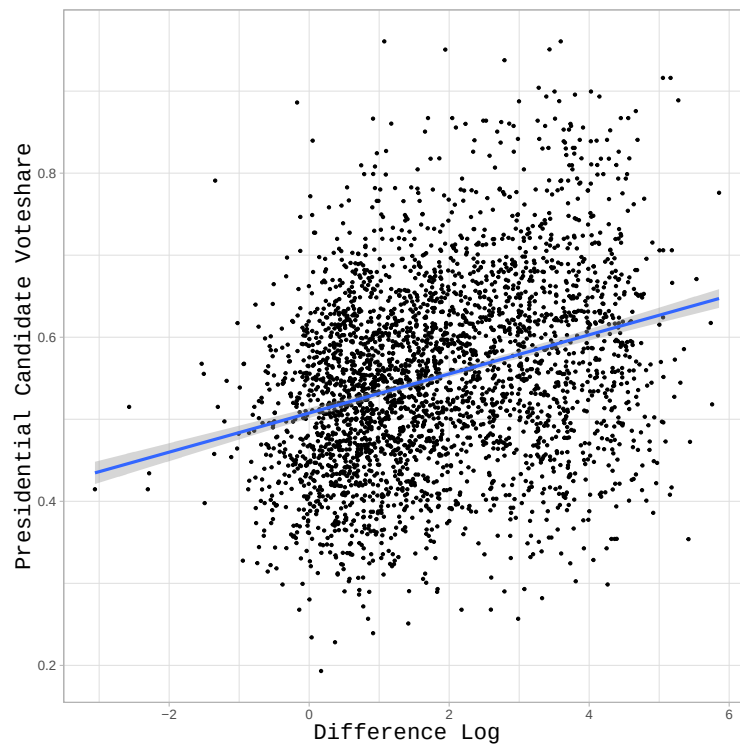
```
1 pdf("PS03_reg2plot.pdf")
2 ggplot(inc.sub, aes(x = difflog, y = presvote)) +
```

```

3 geom_point(size = 0.7) +
4 geom_smooth(method=lm) +
5 labs(x = "Difference Log", y = "Presidential Candidate Voteshare") +
6 theme_light() +
7 theme(axis.title = element_text(size = 15, family = "mono"))
8 dev.off()

```

Figure 2: Presidential Voteshare vs Difflog



3. Save the residuals of the model in a separate object.

```

1 reg2_residuals <- reg2$residuals

```

4. Write the prediction equation.

$$\widehat{\text{presvote}} = 0.508 + 0.024 \times \text{difflog} \quad (2)$$

On average, a one unit increase in difflog is associated with an 0.024 increase in presidential voteshare of incumbent's party.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is **voteshare** and the explanatory variable is **presvote**.

```
1 reg3 <- lm(data= inc.sub, voteshare ~ presvote)
2 summary(reg3)
3 stargazer(reg3,
4           type = "latex",
5           title = "Presidential Candidate Voteshare of the Incumbent
6           Party
7           on the Incumbent's Voteshare",
8           covariate.labels = "Presidential Candidate Voteshare",
9           dep.var.labels = "Voteshare")
```

Table 3: Presidential Candidate Voteshare of the Incumbent Party on the Incumbent's Voteshare

	<i>Dependent variable:</i>
	Voteshare
Presidential Candidate Voteshare	0.388*** (0.013)
Constant	0.441*** (0.008)
Observations	3,193
R ²	0.206
Adjusted R ²	0.206
Residual Std. Error	0.088 (df = 3191)
F Statistic	826.950*** (df = 1; 3191)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

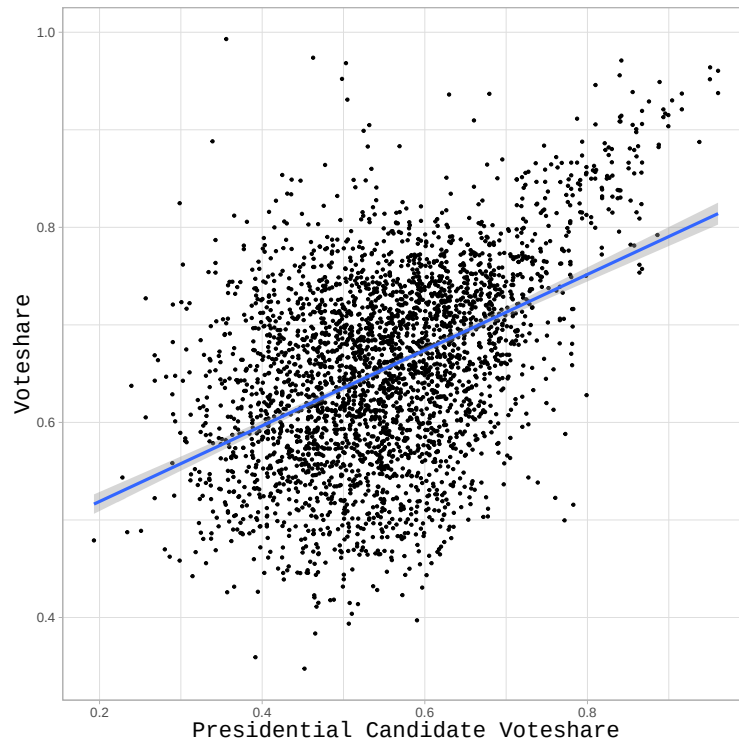
```
1 pdf("PS03_reg3plot.pdf")
2 ggplot(inc.sub, aes(x = presvote, y = voteshare)) +
3   geom_point(size = 0.7) +
4   geom_smooth(method=lm) +
```

```

5 labs(x = "Presidential Candidate Voteshare", y = "Voteshare") +
6 theme_light() +
7 theme(axis.title = element_text(size = 15, family = "mono"))
8 dev.off()

```

Figure 3: Voteshare vs Presidential Voteshare



3. Write the prediction equation.

$$\widehat{\text{voteshare}} = 0.441 + 0.388 \times \text{presvote} \quad (3)$$

On average, a one unit increase in presidential voteshare of incumbent's party is associated with an 0.388 increase in the incumbent's voteshare.

Question 4

The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
1 df$residuals <- data.frame(reg1_residuals, reg2_residuals) #create new df
2 reg4 <- lm(data = df$residuals, reg1_residuals ~ reg2_residuals)
3 summary(reg4)
4 stargazer(reg4,
5           type = "latex",
6           title = "Regression 1 Residuals on Regression 2 Residuals",
7           covariate.labels = "Regression 2 Residuals",
8           dep.var.labels = "Regression 1 Residuals")
```

Table 4: Regression 1 Residuals on Regression 2 Residuals

	<i>Dependent variable:</i>
	Regression 1 Residuals
Regression 2 Residuals	0.257*** (0.012)
Constant	-0.000 (0.001)
Observations	3,193
R ²	0.130
Adjusted R ²	0.130
Residual Std. Error	0.073 (df = 3191)
F Statistic	476.975*** (df = 1; 3191)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two residuals and add the regression line.

```
1 pdf("PS03_reg4plot.pdf")
2 ggplot(df$residuals, aes(x = reg2_residuals, y = reg1_residuals)) +
3   geom_point(size = 0.7) +
4   geom_smooth(method=lm) +
```

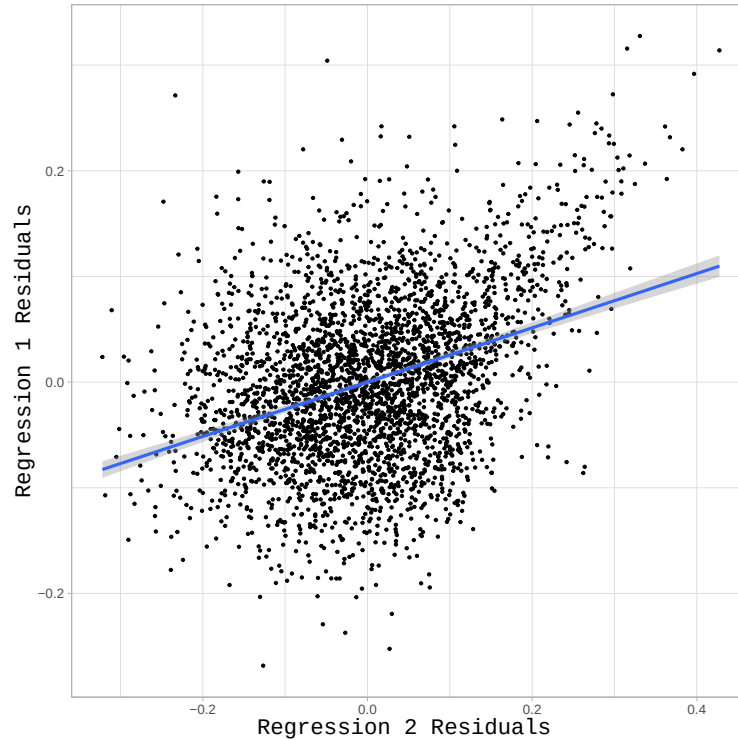


```

5 labs(x = "Regression 2 Residuals ", y = "Regression 1 Residuals") +
6 theme_light() +
7 theme(axis_title = element_text(size = 15, family = "mono"))
8 dev.off()

```

Figure 4: Regression 1 Residuals vs Regression 2 Residuals



3. Write the prediction equation.

$$\widehat{\text{reg1.residuals}} = -0.000 + 0.257 \times \text{reg2.residuals} \quad (4)$$

On average, a one unit increase in Regression 2 residuals is associated with an 0.257 increase in Regression 1 Residuals.

That is, an increase in how much presvote variation is unexplained by difflog is associated with an 0.257 increase in how much voteshare variation is unexplained by difflog.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
1 reg5 <- lm(data = inc.sub, voteshare ~ difflog + presvote) #multivariate
2 summary(reg5)
3 stargazer(reg5,
4           type = "latex",
5           title = "Incumbent/Challenger Difference in Campaign Spending and
6                 the Presidential Candidate Voteshare of the Incumbent
7                 Party
8                 on the Incumbent's Voteshare",
9           covariate.labels = c("Difference Log", "Presidential Candidate
10                             Voteshare"),
11           dep.var.labels = "Voteshare")
```

Table 5: Incumbent/Challenger Difference in Campaign Spending and the Presidential Candidate Voteshare of the Incumbent Party on the Incumbent's Voteshare

	<i>Dependent variable:</i>
	Voteshare
Difference Log	0.036*** (0.001)
Presidential Candidate Voteshare	0.257*** (0.012)
Constant	0.449*** (0.006)
Observations	3,193
R ²	0.450
Adjusted R ²	0.449
Residual Std. Error	0.073 (df = 3190)
F Statistic	1,302.947*** (df = 2; 3190)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

2. Write the prediction equation.

$$\widehat{\text{voteshare}} = 0.449 + 0.036 \times \text{difflog} + 0.257 \times \text{presvote} \quad (5)$$

On average, a one unit increase in difflog is associated with a 0.036 increase in voteshare, holding all else constant. A one unit increase in presvote is, on average, associated with a 0.257 increase in voteshare, holding all else constant.

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

- The regression coefficient of Presidential Vote share (0.257) in model 5 is the same as the regression coefficient of Regression 2 residuals in model 4.
- Model 4 regresses the reg1residuals (i.e. the part of voteshare unexplained by difflog) on reg2residuals (i.e. the part of Presidential voteshare unexplained by difflog).
- Using the residuals that remove the effect of difflog on voteshare and on presvote, model 4 eliminates the confounding effect of difflog on both and computes the independent effect of presidential voteshare on voteshare.
- This is ultimately the same process that occurs when running a multivariate regression, which statistically controls for the effect of other variables in the model.
- In the multivariate model 5 the coefficient of presvote (0.257) is determined independent of the effect of the other explanatory variable difflog, and is therefore identical to that in model 4.